Attributing methane growth with the 2019 budget

Reading note on He et al. (2026)

Attributing 2019–2024 methane growth using TROPOMI satellite observations. Science Advances 12, eadz9007. doi:10.1126/sciadv.adz9007. The published article and supplementary Figs. S1–S9 were obtained.

Method. The authors assimilate 610 million bias-corrected TROPOMI retrievals over 2019–2024 with GEOS-Chem 14.4.1 at $2^\circ \times 2.5^\circ$ resolution, driven by GEOS-FP meteorology. Each annual analytical inversion optimizes 4111 land surface fluxes and two hemispheric OH scaling factors. The preceding posterior becomes the next prior, but prior error variances are reset; 2019 is inverted twice as a burn-in. The base prior errors are 50% for fluxes and 10% for OH, with regularization 0.02. A 27-member ensemble varies these assumptions. The prior combines EDGARv7 and regional anthropogenic inventories, LPJ-MERRA2 wetlands, and MeMo soil uptake. Sector attribution uses the prior sector fractions within each grid cell, rather than independent sector-specific observation constraints.

Budget and findings. Their equation (1) approximates annual methane mass growth as

$$G_j = (G_0 - k_0 \Delta m_j) + \Delta E_j - m_0 \Delta k_{\mathrm{OH},j}, \tag{1}$$

in which 0 denotes 2019, m is mass, E is gross emissions, and k_0 includes OH and other chemical loss. Soil uptake is a separate baseline loss contained in G_0. Changes in other chemical loss, soil uptake, and the product of burden and loss-frequency anomalies are neglected. Emissions rise from 571 Tg yr^{-1} in 2019 to 601 in 2021, then return to 571–575 in 2022–2024. For 2020–2021, 81% of the forcing beyond the baseline relaxation is assigned to emissions and 19% to OH. The 59/25/16% relaxation/emissions/OH partition applies to 2019–2024 and must not be relabeled as a 2019–2022 result. Mean OH lifetime is 11.1 years; OH decreases about 2% from 2019 to 2022. The largest 2019–2024 regional trends are East Africa, South America, and Europe.

Interpretation and reproducibility. The supplement documents fit errors, averaging kernels, sector overlap, error correlations, and sensitivity to regularization, but contains no annual numerical posterior table. The availability statement links model code and satellite inputs, not the numerical posterior products needed to redraw maps or diagnostic matrices. Those inputs do not remove the need for an inversion, which is excluded here. Figure digitization, if used, must be identified as approximate and kept distinct from original numerical data. The livestock/wetland attribution is particularly sensitive to prior spatial overlap and missing tropical observations. The authors note that precipitation-driven wetland maps can miss river inundation; their African ammonia comparison does not support rising livestock. Our decision is to implement their annual decomposition exactly, expose its neglected terms, and distinguish it from a fully integrated constant-2019 counterfactual, whose burden differs from the observed one.